

Practical 4

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Batch: A3

```
import time
```

```
import random
```

```
import json
```

```
from datetime import datetime
```

```
# Simulated ThingSpeak config
```

```
CHANNEL_KEY = "HJEQMRUDGXPTXNPY"
```

```
CHANNEL_ID = "12345"
```

```
SLEEP_INTERVAL = 5
```

```
# Store published data locally (simulates cloud dashboard)
```

```
dashboard = []
```

```
def read_sensor():
```

```
    """Simulate DHT11 sensor readings"""
```

```
    temperature = round(random.uniform(24, 38), 1)
```

```
    humidity = round(random.uniform(50, 85), 1)
```

```
    return temperature, humidity
```

```
def publish_to_cloud(temperature, humidity, count):
```

```
    """Simulate posting to ThingSpeak API"""
```

```
    payload = {
```

```
        "field1": temperature,
```

```
"field2": humidity,  
"key": CHANNEL_KEY  
}
```

```
# Simulate HTTP POST response
```

```
status = 200
```

```
reason = "OK"
```

```
entry_id = count
```

```
record = {  
    "entry_id": entry_id,  
    "timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),  
    "temperature": temperature,  
    "humidity": humidity,  
    "status": status  
}
```

```
dashboard.append(record)
```

```
return status, reason, entry_id
```

```
def print_dashboard():
```

```
    """Print a mini cloud dashboard"""
```

```
    print("\n" + "=" * 55)
```

```
    print("  THINGSPEAK CLOUD DASHBOARD")
```

```
    print(f"    Channel ID : {CHANNEL_ID} | Key: {CHANNEL_KEY[:6]}***")
```

```
    print("=" * 55)
```

```
    print(f"    {'#':<5} {'Timestamp':<22} {'Temp(°C)':<12} {'Humidity(%)}'")
```

```
    print("-" * 55)
```

```
    for r in dashboard[-5:]: # Show last 5 records
```

```
print(f" {r['entry_id']:<5} {r['timestamp']:<22} {r['temperature']:<12} {r['humidity']}")  
print("=" * 55)
```

```
print("IoT Cloud Dashboard Simulation")  
print(f"Publishing to ThingSpeak every {SLEEP_INTERVAL} seconds...")  
print("Press Ctrl+C to stop.\n")
```

```
count = 1
```

```
try:
```

```
    while True:
```

```
        temperature, humidity = read_sensor()
```

```
        print(f"[Sensor Read] Temp: {temperature}°C | Humidity: {humidity}%")
```

```
    try:
```

```
        status, reason, entry_id = publish_to_cloud(temperature, humidity, count)
```

```
        print(f"[HTTP POST] Status: {status} {reason} | Entry ID: {entry_id}")
```

```
    except Exception:
```

```
        print("[HTTP POST] Connection failed")
```

```
print_dashboard()
```

```
count += 1
```

```
time.sleep(SLEEP_INTERVAL)
```

```
except KeyboardInterrupt:
```

```
    print("\n\nFinal Dashboard Data (JSON):")
```

```
    print(json.dumps(dashboard, indent=2))
```

```
    print("\nSimulation stopped.")
```

Output

IoT Cloud Dashboard Simulation

Publishing to ThingSpeak every 5 seconds...

Press Ctrl+C to stop.

[Sensor Read] Temp: 31.1°C | Humidity: 64.0%

[HTTP POST] Status: 200 OK | Entry ID: 1

=====

 THINGSPEAK CLOUD DASHBOARD

Channel ID : 12345 | Key: HJEQMR***

=====

#	Timestamp	Temp(°C)	Humidity(%)
---	-----------	----------	--------------

1	2026-05-09 16:59:58	31.1	64.0
---	---------------------	------	------

=====

[Sensor Read] Temp: 37.7°C | Humidity: 57.8%

[HTTP POST] Status: 200 OK | Entry ID: 2

=====

 THINGSPEAK CLOUD DASHBOARD

Channel ID : 12345 | Key: HJEQMR***

=====

#	Timestamp	Temp(°C)	Humidity(%)
---	-----------	----------	--------------

1	2026-05-09 16:59:58	31.1	64.0
---	---------------------	------	------

2	2026-05-09 17:00:03	37.7	57.8
---	---------------------	------	------

=====

[Sensor Read] Temp: 24.4°C | Humidity: 50.2%

[HTTP POST] Status: 200 OK | Entry ID: 3

=====

 THINGSPEAK CLOUD DASHBOARD

Channel ID : 12345 | Key: HJEQMR***

=====

#	Timestamp	Temp(°C)	Humidity(%)

1	2026-05-09 16:59:58	31.1	64.0
2	2026-05-09 17:00:03	37.7	57.8
3	2026-05-09 17:00:08	24.4	50.2

=====